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DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

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Artificial Neural Network **(LAB)**

Final Project Proposal

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Project Title: Flask-Based AI Customer Service Chatbot for Pizza Outlet

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1.Objective:

The objective of this project is to develop an AI-based customer service chatbot for a pizza outlet using Artificial Neural Networks (ANN) and Flask framework. The chatbot will help customers by answering their queries, showing menu information, taking pizza orders, and providing delivery details automatically. The system is designed to improve customer service and reduce manual workload.

2.Introduction:

Nowadays, many businesses are moving towards automated systems to improve communication with customers. In pizza outlets, employees often spend a lot of time answering repeated customer questions related to menu items, prices, delivery timing, and order placement.

To solve this problem, this project presents a Flask-based AI chatbot application that can interact with customers in real time. The chatbot uses Natural Language Processing (NLP) and Artificial Neural Networks (ANN) to understand customer messages and provide suitable responses.

The application will run through a web interface where users can communicate with the chatbot easily. This system will help improve response speed, customer satisfaction, and service efficiency.

3. Problem Statement:

Traditional customer service systems in pizza restaurants are mostly manual and require human staff for handling customer queries and orders. During busy hours, customers may face delays in getting responses or placing orders.

The main problems include:

- Slow customer support
- Long waiting times
- Repeated manual work
- Difficulty handling multiple customers at the same time

An automated chatbot system is required to improve customer interaction and provide quick responses.

4. Proposed Application:

The proposed system is a web-based chatbot application developed using Flask and ANN. The chatbot will communicate with users through a simple web interface.

The application will:

- Understand customer queries
- Display pizza menu information
- Take customer orders
- Answer frequently asked questions
- Provide delivery information
- Respond automatically in real time

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5. Scope of the Project:

The chatbot application will perform the following tasks:

- Handle greetings and basic conversations
- Show pizza menu with prices
- Help users place orders
- Respond to FAQs
- Provide delivery information
- Generate automated responses
- Improve chatbot performance through updated training data

6. Tools & Technologies:

Programming Language

- Python

Frontend

- HTML
- CSS
- Bootstrap

Backend

- Flask

Libraries & Frameworks

- TensorFlow / Keras
- NLTK
- NumPy
- Pickle

Development Tools

- VS Code
- Jupyter Notebook
- Anaconda

Deployment Platforms

- GitHub
- Render / Railway / PythonAnywhere

7.Dataset:

The dataset used in this project will contain conversational data in JSON format. The dataset will include:

- User queries
- Intent categories
- Predefined chatbot responses

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Some example intents are:

- Greeting
- Pizza order
- Menu inquiry
- Delivery information
- Goodbye

The main dataset file will be intents.json. Additional datasets from Kaggle may also be used to improve chatbot accuracy.

8. Proposed System Architecture:

The chatbot system will follow the following workflow:

1. User enters message on website
2. Flask application receives input
3. Text preprocessing is performed using NLP
4. Feature extraction is applied using Bag of Words
5. ANN model predicts user intent
6. Appropriate response is selected
7. Response is displayed to the user

9. Methodology:

9.1 Data Collection

The chatbot dataset will be collected from custom conversational data and Kaggle resources. Different examples of customer queries will be added for training purposes.

9.2 Data Preprocessing

The text data will be processed using NLTK techniques such as:

- Tokenization
- Lowercasing
- Stopword removal
- Lemmatization

9.3 Feature Extraction

Bag of Words (BoW) technique will be used to convert text into numerical vectors so that the ANN model can process the data.

9.4 Model Design (ANN)

The ANN model will be developed using TensorFlow and Keras.

The model will contain:

- Input Layer
- Hidden Layers with ReLU activation
- Output Layer with Softmax activation

9.5 Model Training

The model will be trained using labeled intent data.

Training details:

- Optimizer: Adam
- Loss Function: Categorical Crossentropy
- Evaluation Metric: Accuracy

9.6 Intent Recognition

The trained ANN model will identify the intent of the user query and classify it into the correct category.

9.7 Response Generation

After intent prediction, the chatbot will generate a suitable response from the predefined dataset.

9.8 Flask Integration

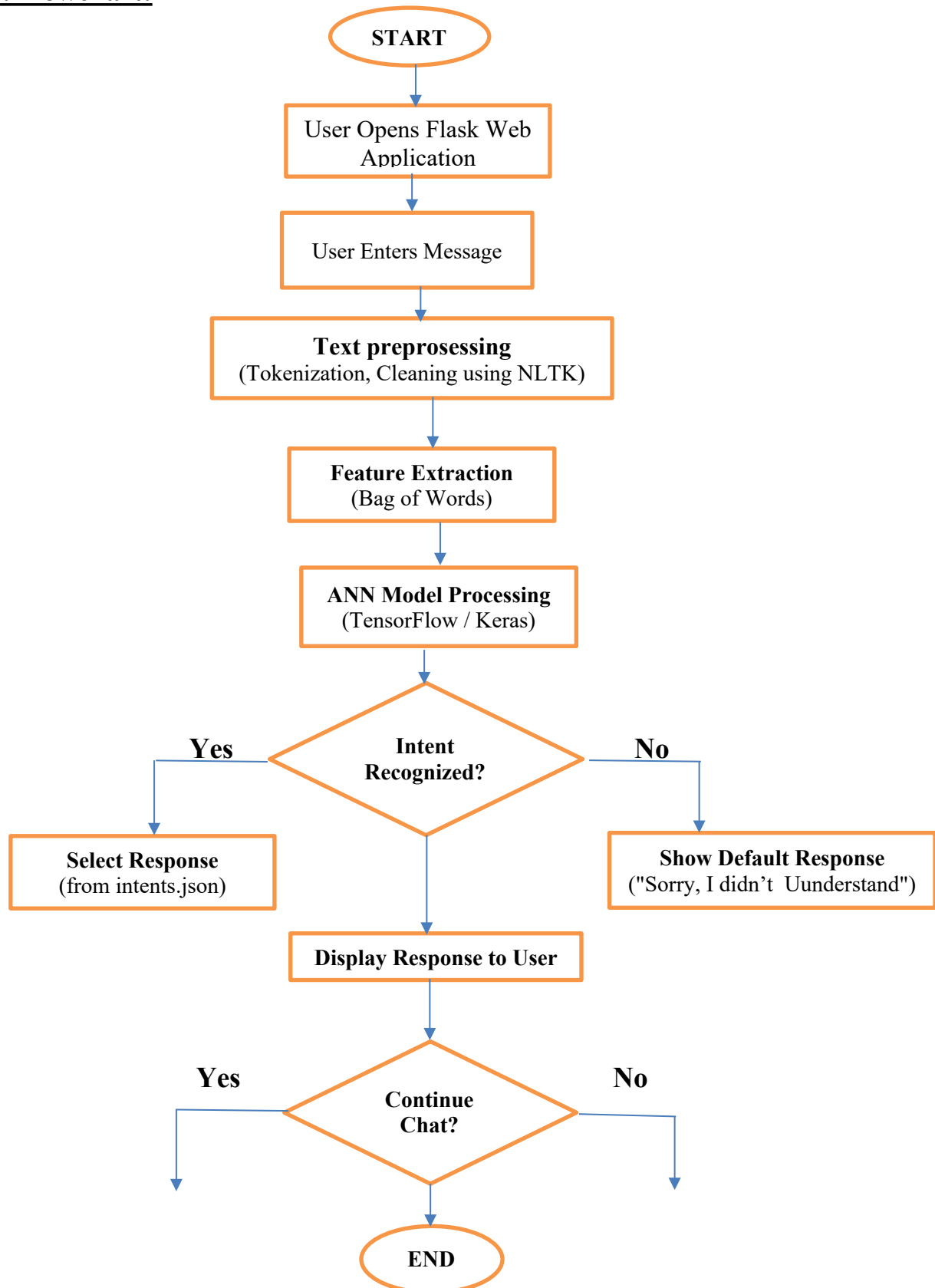
The trained chatbot model will be integrated with the Flask backend to create a working web application.

9.9 Testing & Evaluation

The chatbot will be tested using different user inputs to evaluate:

- Intent recognition accuracy
- Quality of responses
- Application performance
- User interaction

10. Flowchart:



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11. Application Modules:

User Module

- Chat with chatbot
- View pizza menu
- Place orders
- Get delivery information

Admin Module

- Update intents dataset
- Manage chatbot responses
- Improve training data

12. Evaluation & Results:

The performance of the chatbot will be evaluated using:

- Accuracy Graph
- Loss Graph
- Confusion Matrix
- User testing results

13. Deployment:

The Flask chatbot application will be uploaded to GitHub and deployed online using platforms such as Render or Railway. The deployed application link will allow users to access the chatbot through a web browser.

14.Expected Outcome:

The final system will be a fully functional AI chatbot application capable of:

- Providing fast customer support
- Handling pizza-related queries
- Taking customer orders
- Generating intelligent responses
- Improving user experience

15. Limitations:

- Cannot handle highly complex conversations
- Depends on training dataset quality
- Limited understanding of slang words
- Basic version without real-time payment integration

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16. Future Enhancements:

Future improvements may include:

- Voice-based chatbot
- Online payment integration
- Database connectivity
- Mobile application version
- Urdu language support
- Advanced deep learning models such as LSTM and Transformers

15. Conclusion:

This project demonstrates the implementation of Artificial Neural Networks in a real-world Flask application. The chatbot system helps automate customer service operations for pizza outlets and improves communication efficiency. The project also shows how AI can be integrated into web applications to provide smart and interactive services.

16. References:

1. Kaggle – <https://www.kaggle.com/>
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3. Keras – <https://keras.io/>
4. NLTK – <https://www.nltk.org/>
5. Flask – <https://flask.palletsprojects.com/>